

Towards Robust Detection of Chinese Toxic Variants via Dynamic Knowledge Graph–LLM Reasoning

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Abstract

With the growing importance of content safety, toxic language detection, especially in Chinese online environments, has become a key task in natural language processing. However, real-world toxic expressions often appear in obfuscated forms such as pinyin abbreviations, symbol insertion, or visually similar substitutions, making them difficult to detect using traditional rule-based or static models. To address this challenge, we propose a dynamic knowledge graph construction method for toxic text variants, named Variant-KG. This graph encodes diverse structural relations between canonical toxic terms and their variants based on phonetic similarity, visual resemblance, and contextual co-occurrence. A small amount of labeled data is further used to fine-tune large language models (LLMs), enabling initial normalization and variant recognition. On top of this, we design a collaborative detection framework that combines the Variant-KG with frozen LLMs. It performs graph augmented prompting for structure-aware reasoning and adopts a Think-Search-Generate paradigm to dynamically recover broken paths when graph connections are incomplete, enabling both data self-enhancement and knowledge completion during inference. Evaluations on multiple Chinese toxic language datasets show that our model consistently outperforms both non-knowledge-enhanced and existing knowledge-enhanced baselines, demonstrating the effectiveness of our proposed dynamic reasoning framework in handling diverse toxic expressions.

Disclaimer: This paper contains toxic, violent, or discriminatory language examples for research purposes. Some content may be disturbing or offensive to readers. Reader discretion is advised.

CCS Concepts

• Security and privacy → Content filtering; • Computing methodologies → Natural language processing.

Keywords

Toxic language detection, Knowledge graph, Large language models, Graph augmented reasoning, Content safety

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1 Introduction

With the rapid proliferation of social media across the Chinese internet landscape, the dissemination of harmful content has become increasingly covert and diverse [17, 24, 28, 32]. Traditional explicit abusive expressions are now frequently replaced by complex variants such as homophonic substitutions, symbol insertions, and visually similar character manipulations. These representative variants are illustrated in Figure 1, where Vanilla LLMs, lacking prior knowledge of Chinese variant expressions, encounter significant difficulties in accurate identification [8, 25, 26, 29].

Recent advances in content moderation have significantly improved the detection of explicit and canonical toxic expressions. Pretrained language models such as RoBERTa have been widely applied for multilingual toxicity classification [4, 25], while sentiment-aware representations have shown effectiveness in enhancing hate speech detection [14, 39]. Prompt-based techniques leveraging large language models further demonstrate strong adaptability to diverse moderation tasks with limited supervision [6, 9, 34]. In the Chinese context, newly released span-level benchmarks and hateful slang lexicons [2, 28] provide valuable resources for fine-grained detection, addressing linguistic challenges unique to Chinese.

Most existing studies focus on detecting conventional forms of Chinese toxic content, and although some progress has been made, three major challenges remain in terms of both methods and data [3, 10, 31, 33]. First, current approaches largely target canonical toxic expressions while overlooking the rapidly emerging variants in online environments, making rule-based or keyword-matching systems difficult to adapt to continuously evolving manipulations [38]. Second, although pretrained language models exhibit a certain degree of semantic understanding, they often produce high false positive rates on benign content and lack the ability to model the structured relationships between disguised variants and their canonical counterparts [22]. Finally, supervised learning methods are heavily reliant on annotated data, yet the long-tail distribution and fast-evolving nature of Chinese variants make it challenging for labeled datasets to remain timely and comprehensive.

To systematically address the three major challenges in toxic content detection and enhance large language models' capability to

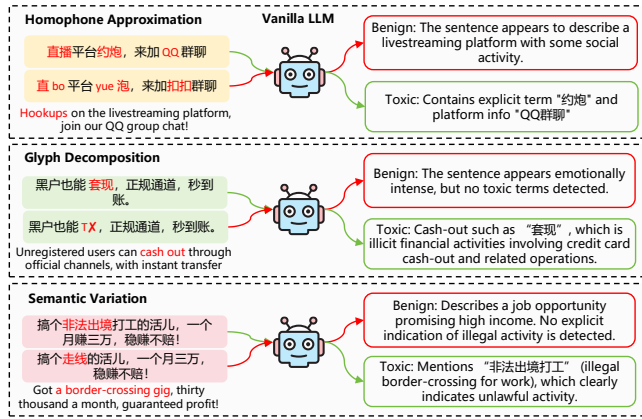


Figure 1: Examples of toxic text variants and detection results across Vanilla and Graph-enhanced LLMs.

recognize Chinese toxic variants, we propose a knowledge graph of Chinese toxic variants, termed **Variant-KG**. This graph is dynamically constructed along the phonetic, visual, and semantic dimensions, covering diverse types of Chinese toxic variants. Building on this, we design a Variant-KG-based detection framework that integrates dynamic knowledge graph construction with collaborative reasoning of LLMs. Unlike prior approaches that rely solely on keyword matching or contextual semantic inference, our framework centers on structure-aware normalization, guiding the dynamic co-evolution of LLMs and the knowledge graph to achieve more accurate and interpretable detection under complex variant perturbations.

At the knowledge representation level, we construct a dynamically evolving heterogeneous knowledge graph that explicitly models multiple types of relationships between canonical toxic terms and their variants, including phonetic substitutions, visual distortions, symbolic insertions, and character-splitting transformations. The graph is initialized through rule-based perturbation modeling and further expanded via LLM-guided triple completion. It also supports dynamic maintenance: edge weights are updated through smoothing to integrate new and historical information, low-confidence edges are pruned, and high-confidence variants discovered during inference are incorporated.

At the semantic reasoning level, we propose a **Think–Search–Generate (TSG)** paradigm to guide LLMs in structure-aware normalization under the support of the knowledge graph. During inference, the model first decomposes the task and generates semantic explanations (Think), then performs path search using graph neighbors (Search), and finally normalizes and classifies the variants (Generate). To enhance robustness, we introduce a graph-augmented prompt construction mechanism, enabling the model to leverage path information when handling out-of-graph variants. In addition, a unified scoring and decision module combines graph confidence with semantic similarity to ensure accuracy and interpretability, thereby improving the model’s generalization to unseen variants.

To thoroughly validate the effectiveness and robustness of the proposed approach, we conduct systematic experiments on our self-constructed high-quality dataset NetSafety-10K as well as two

publicly available datasets. The experiments cover diverse variant perturbation scenarios and provide a comprehensive evaluation of the method. Results demonstrate that Variant-KG consistently outperforms existing baselines across all three datasets, achieving superior performance in recognition accuracy, semantic generalization, and structural transferability.

Our contributions are summarized as follows:

- We propose **Variant-KG**, a novel detection framework for Chinese toxic variants that integrates dynamic knowledge graph construction with structure-aware reasoning of large language models. Unlike prior approaches relying on keyword matching or contextual semantics, Variant-KG achieves more accurate and interpretable detection under complex variant perturbations.
- We introduce a **Think–Search–Generate** reasoning paradigm, where large language models leverage graph-augmented prompts and path information to normalize disguised variants to their canonical counterparts. This mechanism enhances generalization to previously unseen expressions while maintaining interpretability.
- We conduct extensive experiments on three datasets, including our constructed **NetSafety-10K** and two public datasets, demonstrating that Variant-KG consistently outperforms state-of-the-art baselines in recognition accuracy, semantic generalization, and structural transferability.

2 Related Work

Chinese Toxic Detection. Early approaches to Chinese toxic language detection relied heavily on lexicon-based rules and static feature engineering [12, 21, 28]. The rise of deep learning shifted focus to transformer-based models fine-tuned on domain-specific datasets, with benchmarks like COLD [5] and ToxiCN [18] establishing hierarchical taxonomies for offensive language. These works highlighted the granularity needed for nuanced detection but struggled with dynamic, obfuscated variants. To address model robustness, datasets such as ToxiCloakCN [29] were introduced, revealing vulnerabilities of state-of-the-art models to homophonic and emoji perturbations. More recently, large-scale resources such as ChineseHarm-Bench [17] and STATE ToxiCN [2]: have broadened the evaluation scope by covering diverse harmful content categories and safety dimensions. Nevertheless, despite these advances, existing methods and benchmarks still lack flexibility in modeling dynamic toxic variants, leaving them insufficient for real-world adversarial adaptations.

Structured Knowledge Integration. Recent work attempts to address contextual reasoning challenges by integrating structured knowledge. Frameworks like KG-GPT [13] and KICGPT [15, 27] leverage knowledge graphs (KGs) to ground LLM inferences, while Generate-on-Graph [30] employs LLMs as agents for path completion in incomplete KGs. However, these assume static, predefined schemas (e.g., fixed entity-relation triplets), which cannot capture fluid relationships between canonical toxic terms and their variants (e.g., phonetic or visual similarities). Efforts like LightPROF [1] optimize KG retrieval but remain constrained to immutable knowledge sources, lacking adaptability to novel toxic expressions. Hybrid paradigms such as KnowGPT [16, 35] and KoPA [36] align structural

embeddings with LLMs yet focus on schema-complete KGs, neglecting real-world incompleteness in toxicity knowledge. Consequently, static KGs rapidly obsolesce against adversarial innovations, and LLMs hallucinate when reasoning without grounded structural cues [7, 19, 37]. In conclusion, these approaches are overly dependent on static KGs and fail to enable LLMs to recover or augment incomplete or novel toxic knowledge during inference.

3 Problem Definition

We formulate Chinese toxic variant detection as a graph-augmented *normalization-and-classification* task. Let W denote the canonical toxic lexicon and V_t the set of observed variant spans at time t ; the entity set is $\mathcal{V}_t = W \cup V_t$. We maintain a time-evolving knowledge graph G_t whose basic unit is a weighted triple $\langle h, r, t \rangle$, where $r \in \mathcal{R}_t$ is a perturbation relation type (e.g., phonetic, glyph-level, symbol insertion, semantic paraphrase) and each edge has a confidence weight $w_e \in (0, 1)$. For a canonical term $w \in W$ and a variant span $\tilde{v} \in V_t$, the edge confidence is computed by $s_{\text{edge}}(w, \tilde{v})$ and written to the edge weight; a threshold τ controls whether the relation is inserted into G_t .

Given an input text x , we extract candidate spans $\mathcal{E}(x) = \{\tilde{v}\}$ and predict a label $y \in Y$. For each $\tilde{v}$, we retrieve structural evidence from G_t toward candidate canonical terms $w_j \in W$ and obtain an LLM semantic mapping probability $LLM\text{prob}(\tilde{v} \rightarrow w_j)$. By combining structural and semantic evidence, we determine the normalization target w^* and output the category via a mapping $f_{\text{class}} : W \rightarrow Y$ (i.e., $y = f_{\text{class}}(w^*)$), otherwise the instance is classified as normal.

4 Method

To enable systematic modeling and reasoning over Chinese toxic variants, our framework, which is illustrated in Figure 2, is designed as an end-to-end dynamic detection system. It integrates four key modules: (1) an input pre-processing module that identifies potential variant spans in raw text; (2) a **Variant-KG construction** module that encodes multi-type perturbation relations between canonical and obfuscated terms; (3) a **graph evolution** module that continuously updates, smooths, and prunes the knowledge graph to maintain structural reliability; and (4) a **graph-augmented reasoning** module that guides large language models through structure-aware prompts and a **Think-Search-Generate** paradigm for normalization and classification. Together, these components form a closed-loop system that jointly enhances structural completeness, semantic generalization, and detection robustness under evolving variant expressions.

4.1 Variant-KG Construction

Rule-Based Perturbation Modeling. To systematically capture the normalization relationships between canonical and variant expressions, we construct a structured knowledge graph Variant-KG, where each triple $\langle w, r, \tilde{v} \rangle$ denotes a perturbation relation r between a sensitive keyword $w \in \mathcal{W}$ and a variant candidate $\tilde{v} \in \mathcal{V}$. Each edge is assigned a confidence score reflecting the likelihood of normalization under specific linguistic transformations, including phonetic, visual, and semantic shifts.

We model phonetic similarity by converting words into pinyin sequences and computing a normalized edit distance:

$$s_p(w, \tilde{v}) = 1 - \frac{\text{EditDist}_{\text{pinyin}}(w, \tilde{v})}{\max(|\text{pinyin}(w)|, |\text{pinyin}(\tilde{v})|)}. \quad (1)$$

To capture visual variants, we decompose characters into components $P(\cdot)$ and compute the Jaccard similarity:

$$s_g(w, \tilde{v}) = \frac{|P(w) \cap P(\tilde{v})|}{|P(w) \cup P(\tilde{v})|}. \quad (2)$$

For semantic similarity, we use SBERT [23] embeddings and compute contextual similarity based on sampled usage windows:

$$s_c(w, \tilde{v}) = \cos \left(\frac{1}{|C(w)|} \sum_{x \in C(w)} \text{SBERT}(x), \frac{1}{|C(\tilde{v})|} \sum_{y \in C(\tilde{v})} \text{SBERT}(y) \right), \quad (3)$$

where $C(\cdot)$ denotes the contextual sample set for a given token.

Finally, we integrate these signals into a unified edge confidence score using a weighted fusion:

$$s_{\text{edge}}(w, \tilde{v}) = \sigma(\alpha \cdot s_p + \beta \cdot s_g + \gamma \cdot s_c), \quad (4)$$

where $\sigma(\cdot)$ is the sigmoid function and the weights satisfy $\alpha + \beta + \gamma = 1$. A triple is included in the graph $\mathcal{G}$ if $s_{\text{edge}}(w, \tilde{v}) \geq \tau$, where τ is a threshold.

LLM-Guided Triple Completion. While the rule-based Variant-KG effectively captures perturbations such as phonetic and visual variations, many real-world variants originate from complex linguistic phenomena including homophonic puns, metaphors, creative spellings, and contextual cues. These expressions are often highly unstructured and constantly evolving. To address this challenge, we incorporate an LLM as a structural inference engine to support semantic generalization and facilitate dynamic graph evolution (see **Appendix A** for prompt examples).

Formally, given an input sentence x , we query the LLM in inference mode to generate candidate triples:

$$LLM(x) \rightarrow \langle w, r, \tilde{v} \rangle, \quad r \in \mathcal{R} \quad (5)$$

where $\tilde{v}$ is a suspected variant in the input text, w is the inferred canonical form, and r represents the predicted relation type (e.g., homophonic substitution, fuzzy mapping, or semantic implication).

To enhance accuracy and structural consistency, we fine-tune the LLM on a small manually annotated dataset using a triple extraction paradigm. The model is trained to map labeled sentence pairs to triple structures, utilizing parameter-efficient tuning (e.g., LoRA [11]) to preserve most of the frozen model weights. After training, the LLM is used only in inference mode for continuous triple extraction.

For each predicted triple, a confidence score is computed using a unified scoring function $s_{\text{edge}}^{\text{llm}}(w, r, \tilde{v})$, which integrates structural paths, semantic embeddings, and contextual alignment. If the score exceeds a threshold τ_{write} , the edge is appended to the evolving graph:

$$\mathcal{G}_{t+1} = \mathcal{G}_t \cup \left\{ \langle w, r, \tilde{v}, s_{\text{edge}}^{\text{llm}}(w, r, \tilde{v}) \rangle \right\} \quad (6)$$

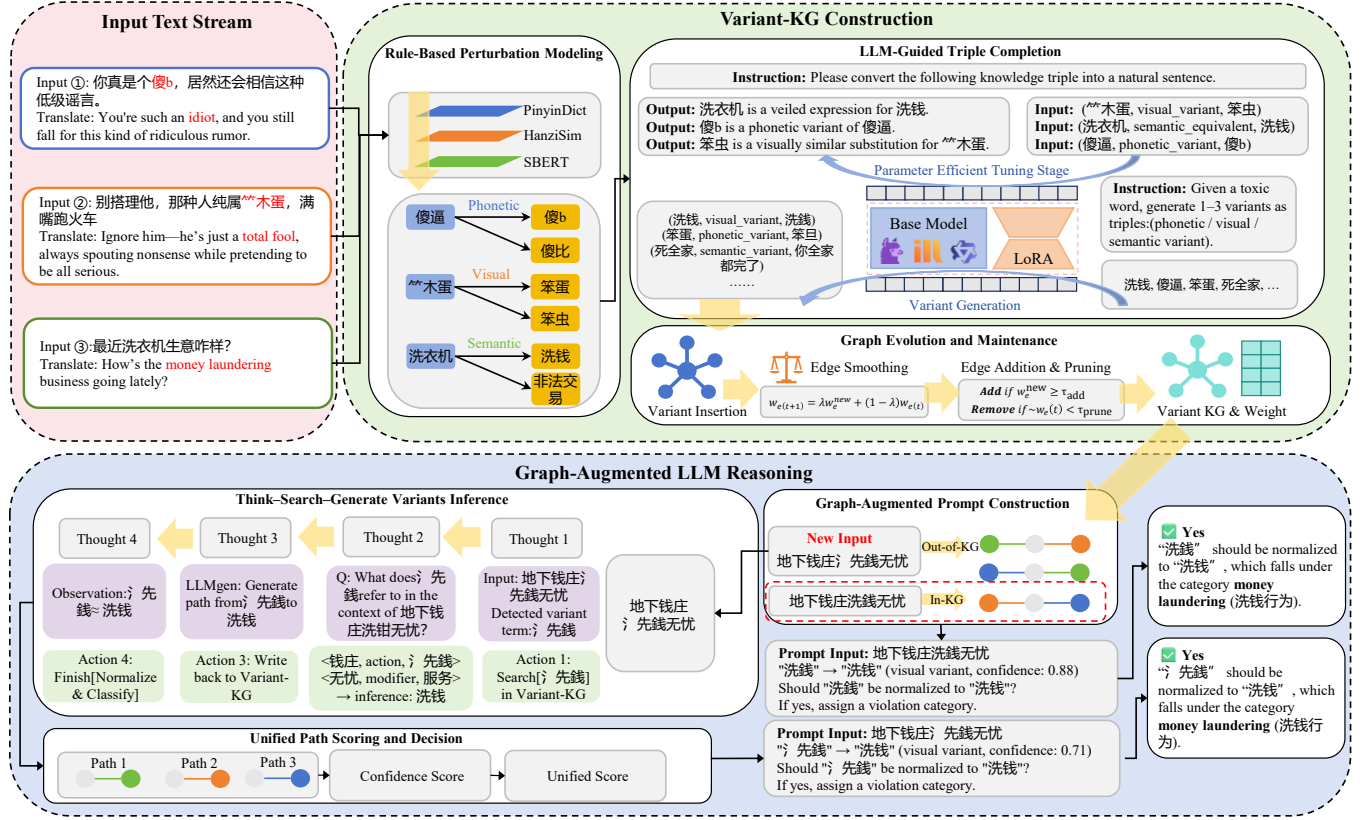


Figure 2: Overview of our framework: Variant-KG Construction builds and maintains the variant graph; Graph-Augmented LLM Reasoning performs normalization and classification with structural guidance.

The newly added edge participates in downstream normalization and detection, and undergoes regular smoothing and pruning to control noise and redundancy. Moreover, the LLM also supports reverse generation: given an existing triple $\langle w, r, \bar{v} \rangle$, it can synthesize natural language samples $\bar{x}$, serving as pseudo-data for augmentation or contrastive training.

Graph Evolution and Maintenance. To accommodate the evolving nature of online language, the Variant-KG is designed to support continuous updates and structural refinement. We introduce a dynamic maintenance mechanism that incrementally incorporates new variant relations while mitigating redundancy and noise.

At each update cycle, newly collected user data within a sliding time window $\mathcal{D}_{\text{inc}}$ are processed by both rule-based and LLM-based extractors to produce an incremental edge set $\mathcal{E}_{\text{inc}}$. For an existing edge e , its confidence weight is refreshed via exponential smoothing:

$$w_e^{t+1} = \lambda w_e^{\text{new}} + (1 - \lambda) w_e^t, \quad (7)$$

where $\lambda \in (0, 1)$ balances adaptability and stability. If a newly detected relation exceeds the confidence threshold τ_{add} , it is appended to the evolving graph:

$$\mathcal{G}_{t+1} = \mathcal{G}_t \cup \{ \langle w, r, \bar{v}, w_e^{\text{new}} \rangle \}. \quad (8)$$

Edges that remain inactive or whose confidence falls below a pruning threshold τ_{prune} are gradually removed, while degree constraints and edge normalization are applied to preserve sparsity and interpretability. This adaptive update mechanism enables the Variant-KG to remain compact, reliable, and continuously aligned with emerging linguistic trends in real-world data.

4.2 Graph-Augmented LLM Reasoning

Although the evolving Variant-KG effectively captures structural relations among variants, it remains insufficient for newly emerging or semantically ambiguous expressions. To address this limitation, we design a **graph-augmented reasoning module** that integrates structure-aware prompting with a **Think-Search-Generate** mechanism, enabling large language models to leverage the Variant-KG for guided reasoning and normalization of unseen or disguised variants.

Graph-Augmented Prompt Construction. Given an input text x , potential variant terms $\bar{v}$ are first identified through rule-based detection. For each detected term not included in the canonical set $\mathcal{W}$, the Variant-KG is queried to retrieve its neighboring canonical nodes $\mathcal{N}(\bar{v})$. If a valid connection $\langle \bar{v}, r, w \rangle$ exists, the most reliable target is selected based on the highest edge confidence:

$$w^* = \arg \max_{w_j \in \mathcal{N}(\bar{v})} w_{ej}. \quad (9)$$

The corresponding relational context and confidence score are then embedded into a structured prompt, serving as external guidance for the language model. By incorporating these graph-derived cues, the model is encouraged to perform structure-aware reasoning—linking suspicious variants to their canonical forms and generating more consistent classification decisions.

Think-Search-Generate Inference. While graph-augmented prompting provides useful structure for normalization, the knowledge graph may still lack direct connections for novel or long-tail variants. Specifically, the confidence score between a variant $\tilde{v}$ and any canonical word $w \in \mathcal{W}$ may fall below a threshold:

$$\forall w \in \mathcal{W}, \text{score}(\tilde{v} \rightsquigarrow w) < \tau_{\text{path}}. \quad (10)$$

To address this gap, we introduce a lightweight Think-Search-Generate inference mechanism, which allows a frozen LLM to complete missing reasoning paths in Variant-KG without requiring fine-tuning.

Given the input context and the neighborhood of $\tilde{v}$, the model first identifies a likely canonical term w using Think-Search-Generate prompting (see **Appendix C** for examples):

$$w_{\text{cand}} = \text{LLM}_{\text{think}}(\tilde{v}, \mathcal{N}(\tilde{v}), \text{prompt}). \quad (11)$$

If w is not directly reachable in the current graph, the model generates a reasoning path to bridge the gap:

$$\text{LLM}_{\text{gen}}(\tilde{v}, w, \mathcal{G}) \rightarrow \left\{ \begin{array}{l} \langle \tilde{v}, r, w \rangle \\ \tilde{v} \rightarrow v^* \rightarrow w \end{array} \right. \quad (12)$$

Each generated edge or multi-hop path is evaluated by the edge confidence function, and if its score exceeds the write threshold τ_{write} , the structure is written back into Variant-KG. This continual update mechanism allows the graph to self-enhance and accumulate reasoning knowledge even under frozen model settings.

Unified Path Scoring and Decision. Given multiple potential normalization paths from a variant $\tilde{v}$ to candidate canonical terms $w_j \in \mathcal{W}$, we integrate both structural evidence and semantic confidence to perform final prediction.

For each candidate path $P_j : \tilde{v} \rightsquigarrow w_j$, we compute a structural confidence score by multiplying the weights of all edges in the path:

$$\text{score}(P_j) = \prod_{e \in P_j} w_e. \quad (13)$$

Simultaneously, the LLM estimates a semantic matching probability $\text{LLM}_{\text{prob}}(\tilde{v} \rightarrow w_j)$, indicating how likely $\tilde{v}$ maps to w_j based on context.

We then combine both signals into a unified score:

$$\begin{aligned} \text{UnifiedScore}(w_j) &= \lambda \cdot \text{score}(P_j) \\ &+ (1 - \lambda) \cdot \text{LLM}_{\text{prob}}(\tilde{v} \rightarrow w_j) \end{aligned} \quad (14)$$

The final normalization target is selected as:

$$w^* = \arg \max_{w_j \in \mathcal{W}} \text{UnifiedScore}(w_j). \quad (15)$$

If the unified score of w^* exceeds a predefined threshold, $\tilde{v}$ is normalized to w^* and assigned the corresponding violation category. This result is further written back into the graph as a new triple $\langle \tilde{v}, r, w^* \rangle$, facilitating future inference.

Table 1: Variant budgets in the NetSafety-10K dataset. ER denotes the edit ratio on toxic spans; Tok/Sent represent the maximum number of edits per token/sentence; Sim denotes semantic similarity.

Variant Type	ER (%)	Tok	Span	Sent	Sim
Homophone / Phonetic (HOMO)	≤ 30	≤ 1	$\leq 2; \text{len} \pm 20\%$	≤ 6	–
Glyph / Visual (GLYPH)	≤ 25	≤ 1	+2 ZWSP	≤ 6	–
Symbol / Insertion (SYMB)	≤ 20	≤ 2	–	≤ 5	–
Semantic / Euphemism (SEMA)	$\leq 30^*$	–	≤ 2 phr.	–	≥ 0.85

Finally, if any variant $\tilde{v}$ in the input text is successfully normalized to a canonical term with a valid violation label, the input is flagged as toxic, and the associated category is output as the detection result.

5 Experiment

5.1 Performance Evaluation

Dataset. (1) ToxiCN [2] and (2) ChineseHarm-Bench [17], both standard Chinese, non-variant corpora (fine-grained labels from Zhihu/Tieba and a professionally annotated benchmark with a knowledge rule base, respectively); and (3) NetSafety-10K, a variant-rich corpus collected from major Chinese online platforms. In addition to standard violation labels such as *Political Sensitivity*, *Vulgar/Pornographic*, *Public Disorder*, *Abusive Language*, *Feudal Superstition*, *Illegal Activity*, *Underground Advertising*, and *Others*, NetSafety-10K explicitly incorporates four adversarial variant types: *phonetic*, *glyph/visual*, *symbol/insertion*, and *euphemism*, for comprehensive robustness evaluation (see Table 1).

Model Backbones and Variants. We evaluate four backbone models: GPT-4o Mini, Llama-3-8B-Instruct, Qwen3-8B, and Baichuan-7B, to test the generalizability of knowledge-augmented reasoning. For each backbone, we report two variants: (1) a *vanilla* version (prompt-only, without fine-tuning or KG integration) and (2) a *KG-enhanced* version that integrates our Variant-KG. The open-source models are fine-tuned on a toxic lexicon using parameter-efficient adaptation, while GPT-4o Mini remains frozen. Implementation details, including LoRA configuration, decoding temperature, output length, and KG pruning, are provided in the Implementation Details section. This setup enables an apples-to-apples comparison of Variant-KG across different model families.

Baselines. We compare against three families of baselines: (1) *BERT-style classifiers*: a Chinese BERT-base model fine-tuned on our training split, and COLDETECTOR [5]. (2) *LLMs (prompt-only, few-shot)*: GPT-4o Mini, Llama-3-8B-Instruct, Qwen3-8B, and Baichuan-7B evaluated under few-shot prompting to probe standalone generalization; these models perform well on explicit toxicity but degrade notably under adversarial variants. (3) *Knowledge-enhanced methods*: C²TU [20], a rule-driven detector leveraging a homophone-aware toxic lexicon and handcrafted rules; and our Variant-KG integrated into different backbones to test architecture-agnostic improvements.

Table 2: Acc on clean and adversarial settings of NetSafety-10K (mean over four variant types). Higher is better.

Method	Canonical	Variant
BERT-base (fine-tuned)	81.2%	73.4%
COLDETECTOR (off-the-shelf)	51.3%	42.5%
GPT-4o Mini (few-shot)	82.9%	66.0%
Llama-3-8B (few-shot)	59.7%	59.0%
Qwen3-8B (few-shot)	68.2%	63.7%
Baichuan-7B (few-shot)	67.5%	63.3%
C ² TU (rule-based)	80.1%	71.0%
Llama-3-8B + Variant-KG	89.0%	85.0%
Baichuan-7B + Variant-KG	88.5%	84.0%
Ours (Qwen3-8B + Variant-KG)	90.8%	88.2%

Implementation Details. We construct the Variant-KG by deduplicating a toxic lexicon, generating phonetic variants via pinyin rewrites, visual variants through character-shape substitutions, and semantic variants using Gemini. KG pruning is performed with a decoding temperature of 0.4, reasoning with 0.1, and KG-augmented prompts are limited to 1,024 tokens. From the KG, we synthesize 8K question-answer pairs, where each question queries the relation between two variants and each answer provides Gemini’s explanation (split 95/5 for training and validation). We fine-tune each model using LoRA (targeting $q/k/v/o$ matrices; $r=8$, dropout = 0.1) with AdamW optimizer (learning rate 5×10^{-5} , weight decay 0.05, warm-up 0.06, one epoch, batch size 1, gradient accumulation 8, and gradient checkpointing). Prompts are left-truncated to preserve the full answer. During inference, the model extracts suspected variants with confidence scores, retrieves the top- k ($= 5$) neighbors from the KG based on cosine similarity over frozen sentence embeddings, and combines the scores through a fixed weighted rule.

Results. Table 2 compares detectors under canonical (no attack) and variant-perturbed settings. Specifically, we split NetSafety-10K into a *Canonical* subset and a *Variant* subset generated by four perturbation types. Our Variant-KG consistently outperforms all baselines in both regimes and exhibits the smallest Canonical-to-Variant drop, indicating that it preserves accuracy when surface forms are adversarially perturbed rather than overfitting to canonical patterns. Table 3 further reports Acc/Macro-F1/FPR on CH-B, NetSafety-10K, and ToxiCN across multiple backbones (Llama-3, Qwen, Baichuan). Although the task is multi-class, we compute FPR by collapsing labels into *violation* vs. *non-violation*, so that false alarms are measured consistently across datasets. Across datasets and backbones, Variant-KG improves Macro-F1 while keeping FPR low, suggesting better recall on minority violation classes without sacrificing precision on benign content. The gains are most pronounced on the Variant subset and under stronger perturbations, supporting the robustness of KG-grounded evidence fusion and demonstrating architecture-agnostic generalizability.

5.2 Component Analysis

We now examine the contribution of each component in our variant-aware detector. The full system integrates five modules: a Variant-KG with four edge types, KG-grounded QA supervision for the

extractor, retrieval + fusion of model confidence with KG evidence, graph pruning, and TSG. To quantify their roles, we ablate or replace one piece at a time, yielding the following configurations: (1) Full Components: our complete pipeline. (2) w/o QA fine-tuning: remove KG-grounded QA supervision; the extractor is not adapted with relation QAs. (3) w/o KG (model-only): disable retrieval and fusion; decisions rely solely on the backbone’s confidence (LoRA-tuned LLM without any KG components). (4) w/o Model Score (KG-Only): Removing the model score and relying solely on KG evidence for decision-making, using nearest neighbors and thresholds. (5) w/o HOMO Edge: Removing the phonetic relation (HOMO) from the KG to test its effect on the model’s performance. (6) w/o SEMA Edge: Removing the semantic relation (SEMA) from the KG to assess the impact on model performance. (7) w/o pruning: construct KG without pruning/merging, exposing the model to noisier neighbors. (8) w/o TSG: We disable TSG-based path completion when the KG lacks confident links, relying only on existing KG neighbors for graph-augmented prompting.

Table 4 summarizes our ablations. Removing any component reduces Acc, with the largest drop occurring w/o QA fine-tuning, highlighting the importance of relation-aware QA supervision. Among edge-type ablations, removing HOMO or SEMA leads to the biggest performance loss, indicating their key role in detecting evasive toxicity. The w/o KG (Model-Only) and KG-only configurations show the complementary roles of model confidence and KG evidence, as both underperform compared to the full system. w/o Pruning results in only a minor loss, while disabling TSG causes a clear degradation, confirming that path-completion reasoning is crucial when the KG lacks confident connections, especially for out-of-KG variants. Overall, the improvements come from the combined effect of QA-trained extraction, KG retrieval/fusion, and TSG-based path completion.

5.3 Performance Analysis

Inference Efficiency Across Methods. We evaluate per-sample latency and throughput on a single RTX 3090 GPU (batch size = 1, identical decoding length across models). The compared methods include Llama-3-8B + KG, Qwen3-8B + KG, Baichuan-7B + KG, and a fine-tuned BERT baseline, with results summarized in Table 5. KG-enhanced LLMs exhibit second-level latency, primarily dominated by decoder generation, whereas the BERT model operates at millisecond-level latency and therefore achieves the highest throughput. The additional KG retrieval step introduces only a minor overhead compared with decoding, owing to efficient indexing and parallel retrieval. This reflects a trade-off: KG integration offers substantially higher robustness against variant attacks but at the cost of lower throughput than lightweight BERT-style models. Latency is also more sensitive to output length and backbone size.

To further optimize efficiency, we prebuild and persist an ANN index that is memory-mapped upon startup to avoid reconstruction. During inference, we perform parallel top- k retrieval using multi-probe ANN search and batched cosine scoring. The average retrieval time is approximately 0.003 s per query, which remains negligible compared with LLM decoding even as the KG expands. In practice, end-to-end latency scales primarily with decoding length and backbone size, rather than the size of the knowledge graph.

Table 3: Accuracy, Macro-F1, and False Positive Rate (FPR) on ChineseHarm-Bench (CH-B), NetSafety-10K, and ToxiCN. Knowledge-enhanced models show consistent gains across multiple backbones; C²TU yields the lowest FPR due to conservative rule coverage.

Model	ChineseHarm-Bench (CH-B)			NetSafety-10K			ToxiCN		
	Acc.	Macro-F1	FPR	Acc.	Macro-F1	FPR	Acc.	Macro-F1	FPR
<i>BERT-based models</i>									
BERT (fine-tuned)	71.9%	65.8%	5.3%	88.0%	80.0%	4.5%	79.4%	72.0%	4.6%
COLDETECTOR	58.3%	54.5%	4.7%	42.5%	39.3%	4.0%	62.0%	58.0%	4.3%
<i>LLMs with prompt tuning</i>									
GPT-4o Mini	70.3%	69.4%	4.0%	67.0%	63.2%	3.5%	76.8%	74.1%	3.7%
Llama3-8B	60.9%	61.6%	3.1%	59.7%	57.4%	2.8%	67.5%	66.2%	2.9%
Qwen3-8B	66.9%	63.4%	3.0%	64.5%	61.2%	2.6%	72.9%	70.4%	2.8%
Baichuan-7B	65.7%	62.5%	2.9%	63.3%	60.3%	2.4%	71.6%	69.5%	2.6%
<i>Knowledge-enhanced methods</i>									
C ² TU (rule-based)	70.0%	64.1%	0.5%	78.0%	71.5%	0.8%	74.5%	69.8%	0.6%
Llama3-8B + KG (Proposed)	85.8%	81.0%	1.9%	93.2%	88.0%	1.2%	87.9%	83.0%	1.7%
Qwen3-8B + KG (Proposed)	87.5%	83.4%	1.8%	94.7%	90.1%	1.1%	89.1%	85.2%	1.5%
Baichuan-7B + KG (Proposed)	86.6%	79.8%	2.0%	91.7%	88.2%	1.3%	87.0%	82.1%	1.8%

Table 4: Component analysis. This table reports Acc on three datasets, showing that removing QA fine-tuning causes the largest drop, while removing HOMO/SEMA edges or write-back further reduces robustness and pruning largely preserves it.

Model / Setting	CH-B	NetSafety-10K	ToxiCN
<i>Ablations aligned with our pipeline (extract → retrieve → fuse → write-back)</i>			
w/o QA fine-tuning	73.0%	74.5%	73.2%
w/o KG (model-only)	85.2%	86.8%	84.8%
w/o Model Score (KG-Only)	64.2%	78.2%	68.6%
w/o HOMO Edges	86.0%	89.1%	86.1%
w/o SEMA Edges	85.4%	88.8%	85.6%
w/o pruning	85.8%	89.4%	86.1%
w/o TSG	85.7%	89.0%	85.8%
Full Components	87.5%	94.7%	89.1%

Hallucination and Reliability. We observe two failure points: (i) *extraction*—the LLM invents a suspected variant not grounded in the input text; and (ii) *write-back*—when proposing new variants for the KG, the LLM occasionally fabricates edges. Figure 3 reports the extraction hallucination rate before/after our controls on the NetSafety-10K dataset.

Extraction. We constrain the extractor to return only input-anchored spans. If no grounded span exists, it must output NONE. In addition, we deploy a high-risk hallucination lexicon curated from 100k offline samples; any candidate that hits this list without strong local context support is rejected. These two controls suppress free-form inventions while keeping recall on genuine homophone/visual disguises.

Write-back (similarity-gated promotion). For each suspected variant in the input and each LLM-generated candidate, we compute

a semantic/orthographic similarity σ between their embeddings and combine it with the model’s calibrated confidence s_m :

$$s = \lambda s_m + (1 - \lambda) \sigma, \quad (16)$$

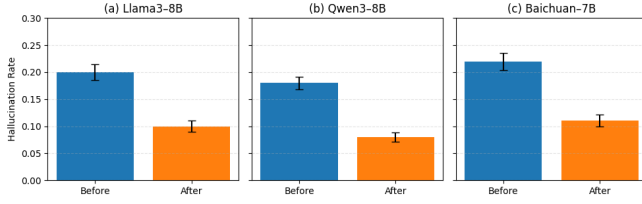
with gating by a confidence threshold τ and a disagreement tolerance δ :

$$\mathbb{I}_{\text{accept}} = \mathbb{I}[s \geq \tau] \cdot \mathbb{I}[|s_m - \sigma| \leq \delta]. \quad (17)$$

Type-wise edit budgets are enforced: HOMO/GLYPH/SYMB must satisfy the predefined edit limits; SEMA further requires $\sigma \geq \sigma_{\min}$. Accepted edges first enter a *shadow KG* for short probation and are promoted only if they do not raise FPR and yield non-negative $\Delta\text{Macro-F1}$ on a held-out block.

Table 5: Inference latency and throughput. Measured on a single RTX 3090 GPU (batch size = 1, $k=5$ retrieval, max output = 64 tokens).

Model	Latency (s)	Throughput (samples/s)
Llama-3-8B + KG	2.80	0.36
Qwen3-8B + KG	2.50	0.40
Baichuan-7B + KG	2.70	0.37
BERT (fine-tuned)	0.012	83.0

**Figure 3: Extraction-stage hallucination rate before vs. after. Bars compare vanilla extraction (*Before*) with our prompting + high-risk lexicon (*After*) for Llama3-8B, Qwen3-8B, and Baichuan-7B. Error bars denote standard error.**

6 Case Study

To further demonstrate the effectiveness of our framework, we present two representative examples from the ChineseHarm-Bench and NetSafety-10K datasets (Figure 4). Each illustrates typical failure cases of standalone LLMs and how our Variant-KG reasoning resolves them.

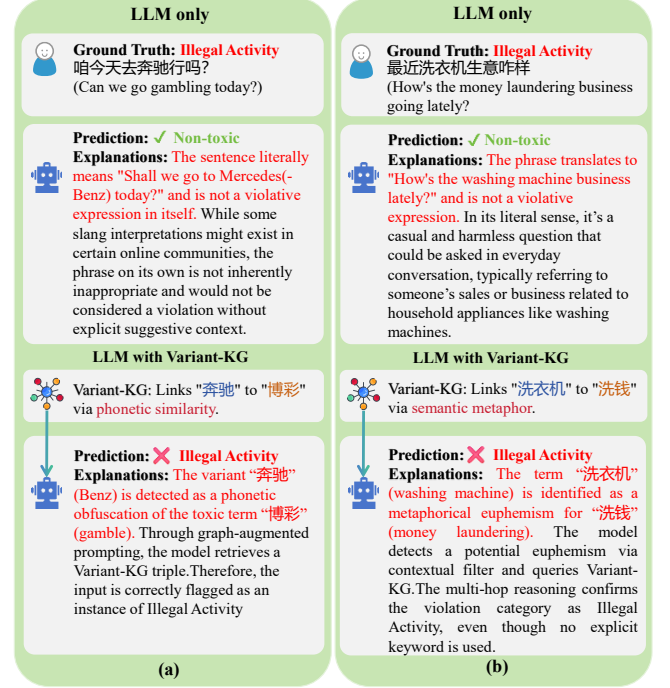
Case (a) contains the phrase “Benchi” (“Benz”), a visually and phonetically disguised form of “Bocai” (“gamble”). The base LLM interprets it literally and classifies the input as non-toxic. In contrast, our Variant-KG framework retrieves the triple (Benchi, phonetic, Bocai) through multi-hop reasoning, enabling correct normalization and classification as *Illegal Activity*. This case highlights the role of phonetic reasoning and structural grounding in uncovering hidden toxic intent.

Case (b) includes the seemingly benign phrase “How’s the washing machine business lately?”, which euphemistically refers to “money laundering.” Without contextual grounding, standard LLMs misclassify it as benign. Our model identifies “washing machine” as a metaphor for “money laundering” via triples like (washing machine, euphemism, money laundering), and verifies it through the Think–Search–Generate reasoning process, correctly labeling it as *Illegal Activity*. This example underscores the importance of contextual and metaphorical reasoning for detecting disguised toxicity.

Additional qualitative examples and visualization analyses are provided in **Appendix F**.

7 Conclusion

In this work, we present a dynamic knowledge graph-LLM reasoning framework, **Variant-KG**, for robust detection of Chinese toxic

**Figure 4: Case study examples from ChineseHarm-Bench and NetSafety-10K.**

variants. By integrating structural cues with language model inference, our method effectively normalizes disguised expressions and adapts to evolving online language. Through extensive experiments across multiple datasets and model architectures, we demonstrate consistent improvements in both accuracy and robustness over strong baselines. Our results highlight the importance of structure-aware reasoning and dynamic knowledge evolution in addressing the challenges of obfuscated toxic content in real-world settings.

Beyond empirical gains, Variant-KG offers broader insights into the synergy between symbolic structures and neural reasoning. The dynamically evolving graph captures the complexity of phonetic, visual, and semantic perturbations, while the proposed **Think–Search–Generate** paradigm enables LLMs to generalize to previously unseen variants in an interpretable manner. Together, these contributions provide a promising foundation for building safer, adaptive, and trustworthy moderation systems in dynamic online environments.

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References

- [1] Tu Ao, Yanhua Yu, Yuling Wang, Yang Deng, Zirui Guo, Liang Pang, Pinghui Wang, Tat-Seng Chua, Xiao Zhang, and Zhen Cai. 2025. Lightprof: A light-weight reasoning framework for large language model on knowledge graph. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 23424–23432.
- [2] Zewen Bai, Shengdi Yin, Junyu Lu, Jingjie Zeng, Haohao Zhu, Yuanyuan Sun, Liang Yang, and Hongfei Lin. 2025. STATE ToxiCN: A Benchmark for Span-level Target-Aware Toxicity Extraction in Chinese Hate Speech Detection. *arXiv preprint arXiv:2501.15451* (2025).
- [3] Irina Bigoulaeva, Harish Tayyar Madabushi, and Iryna Gurevych. 2025. The inherent limits of pretrained llms: The unexpected convergence of instruction tuning and in-context learning capabilities. *arXiv preprint arXiv:2501.08716* (2025).
- [4] Tommaso Caselli, Valerio Basile, Jelena Mitrović, and Michael Granitzer. 2020. HateBERT: Retraining BERT for abusive language detection in English. *arXiv preprint arXiv:2010.12472* (2020).
- [5] Jiawen Deng, Jingyan Zhou, Hao Sun, Chujie Zheng, Fei Mi, Helen Meng, and Minlie Huang. 2022. COLDA: A benchmark for Chinese offensive language detection. *arXiv preprint arXiv:2201.06025* (2022).
- [6] Hongyu Gong, Yuchen Li, Suma Bhat, and Pramod Viswanath. 2019. Context-sensitive malicious spelling error correction. In *The World Wide Web Conference*. 2771–2777.
- [7] Xinyan Guan, Yanjiang Liu, Hongyu Lin, Yaojie Lu, Ben He, Xianpei Han, and Le Sun. 2024. Mitigating large language model hallucinations via autonomous knowledge graph-based retrofitting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 18126–18134.
- [8] Xiaochuang Han and Yulia Tsvetkov. 2020. Fortifying toxic speech detectors against veiled toxicity. *arXiv preprint arXiv:2010.03154* (2020).
- [9] Xinlei He, Savvas Zannettou, Yun Shen, and Yang Zhang. 2024. You only prompt once: On the capabilities of prompt learning on large language models to tackle toxic content. In *2024 IEEE Symposium on Security and Privacy (SP)*. IEEE, 770–787.
- [10] Alec Helbling, Mansi Phute, Matthew Hull, and Duen Horng Chau. 2023. Llm self defense: By self examination, llms know they are being tricked. *arXiv e-prints* (2023), arXiv–2308.
- [11] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. 2022. Lora: Low-rank adaptation of large language models. *ICLR* 1, 2 (2022), 3.
- [12] Aiqi Jiang, Xiaohan Yang, Yang Liu, and Arkaitz Zubiaga. 2022. Swsr: A chinese dataset and lexicon for online sexism detection. *Online Social Networks and Media* 27 (2022), 100182.
- [13] Jiho Kim, Yeonsu Kwon, Yohan Jo, and Edward Choi. 2023. Kg-gpt: A general framework for reasoning on knowledge graphs using large language models. *arXiv preprint arXiv:2310.11220* (2023).
- [14] Alyssa Lees, Daniel Borkan, Ian Kivlichan, Jorge Nario, and Tesh Goyal. 2021. Capturing covertly toxic speech via crowdsourcing. In *Proceedings of the First Workshop on Bridging Human-Computer Interaction and Natural Language Processing*. 14–20.
- [15] Yanlin Li, Ning Chen, Guangrong Zhao, and Yiran Shen. 2024. Kd-eye: Light-weight pupil segmentation for eye tracking on vr headsets via knowledge distillation. In *International Conference on Wireless Artificial Intelligent Computing Systems and Applications*. Springer, 209–220.
- [16] Yanlin Li, Hao Liu, Huimin Liu, Yinwei Wei, and Yupeng Hu. 2025. MIST: Towards Multi-dimensional Implicit Bias and Stereotype Evaluation of LLMs via Theory of Mind. *arXiv preprint arXiv:2506.14161* (2025).
- [17] Kangwei Liu, Siyuan Cheng, Bozhong Tian, Xiaozhuan Liang, Yuyang Yin, Meng Han, Ningyu Zhang, Bryan Hooi, Xi Chen, and Shumin Deng. 2025. ChineseHarm-Bench: A Chinese Harmful Content Detection Benchmark. *arXiv preprint arXiv:2506.10960* (2025).
- [18] Junyu Lu, Bo Xu, Xiaokun Zhang, Changrong Min, Liang Yang, and Hongfei Lin. 2023. Facilitating fine-grained detection of Chinese toxic language: Hierarchical taxonomy, resources, and benchmarks. *arXiv preprint arXiv:2305.04446* (2023).
- [19] Linhao Luo, Zicheng Zhao, Gholamreza Haffari, Yuan-Fang Li, Chen Gong, and Shirui Pan. 2024. Graph-constrained reasoning: Faithful reasoning on knowledge graphs with large language models. *arXiv preprint arXiv:2410.13080* (2024).
- [20] Xuchen Ma, Jianxiang Yu, Wenming Shao, Bo Pang, and Xiang Li. 2025. Breaking the Cloak! Unveiling Chinese Cloaked Toxicity with Homophone Graph and Toxic Lexicon. *arXiv preprint arXiv:2505.22184* (2025).
- [21] David Noever. 2018. Machine learning suites for online toxicity detection. *arXiv preprint arXiv:1810.01869* (2018).
- [22] Nedjma Ousidhoum, Xinran Zhao, Tianqing Fang, Yangqiu Song, and Dit-Yan Yeung. 2021. Probing toxic content in large pre-trained language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. 4262–4274.
- [23] Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084* (2019).
- [24] Anna Schmidt and Michael Wiegand. 2017. A survey on hate speech detection using natural language processing. In *Proceedings of the fifth international workshop on natural language processing for social media*. 1–10.
- [25] Hui Su, Weiwei Shi, Xiaoyu Shen, Zhou Xiao, Tuo Ji, Jiarui Fang, and Jie Zhou. 2022. Rocbert: Robust chinese bert with multimodal contrastive pretraining. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 921–931.
- [26] Yuxia Wang, Zenan Zhai, Haonan Li, Xudong Han, Lizhi Lin, Zhenxuan Zhang, Jingru Zhao, Preslav Nakov, and Timothy Baldwin. 2024. A Chinese dataset for evaluating the safeguards in large language models. *arXiv preprint arXiv:2402.12193* (2024).
- [27] Yanbin Wei, Qiushi Huang, James T Kwok, and Yu Zhang. 2024. Kicgpt: Large language model with knowledge in context for knowledge graph completion. *arXiv preprint arXiv:2402.02389* (2024).
- [28] Yunze Xiao, Houda Bouamor, and Wajdi Zaghouani. 2024. Chinese offensive language detection: current status and future directions. *arXiv preprint arXiv:2403.18314* (2024).
- [29] Yunze Xiao, Yujia Hu, Kenny Tsu Wei Choo, and Roy Ka-wei Lee. 2024. Toxi-cloakcn: Evaluating robustness of offensive language detection in chinese with cloaking perturbations. *arXiv preprint arXiv:2406.12223* (2024).
- [30] Yao Xu, Shizhu He, Jiabei Chen, Zihao Wang, Yangqiu Song, Hanghang Tong, Guang Liu, Kang Liu, and Jun Zhao. 2024. Generate-on-graph: Treat llm as both agent and kg in incomplete knowledge graph question answering. *arXiv preprint arXiv:2404.14741* (2024).
- [31] Hsu Yang and Chuan-Jie Lin. 2020. Tocc: A dataset for chinese profanity processing. In *Proceedings of the Second Workshop on Trolling, Aggression and Cyberbullying*. 6–12.
- [32] Qingpo Yang, Yakai Chen, Zihui Xu, Yu-ming Shang, Sanchuan Guo, and Xi Zhang. 2025. Scdd: A session-based dataset for chinese cyberbullying detection. *arXiv preprint arXiv:2501.15042* (2025).
- [33] Hengxiang Zhang, Hongfu Gao, Qiang Hu, Guanhua Chen, Lili Yang, Bingyi Jing, Hongxin Wei, Bing Wang, Haifeng Bai, and Lei Yang. 2024. Chinesesafe: A chinese benchmark for evaluating safety in large language models. *arXiv preprint arXiv:2410.18491* (2024).
- [34] Min Zhang, Jianfeng He, Taoran Ji, and Chang-Tien Lu. 2024. Don't Go To Extremes: Revealing the Excessive Sensitivity and Calibration Limitations of LLMs in Implicit Hate Speech Detection. *arXiv preprint arXiv:2402.11406* (2024).
- [35] Qinggang Zhang, Junnan Dong, Hao Chen, Daochen Zha, Zailiang Yu, and Xiao Huang. 2024. Knowgpt: Knowledge graph based prompting for large language models. *Advances in Neural Information Processing Systems* 37 (2024), 6052–6080.
- [36] Yichi Zhang, Zhuo Chen, Lingbing Guo, Yajing Xu, Wen Zhang, and Huajun Chen. 2024. Making large language models perform better in knowledge graph completion. In *Proceedings of the 32nd ACM international conference on multimedia*. 233–242.
- [37] Yibo Zhao, Jiapeng Zhu, Can Xu, Yao Liu, and Xiang Li. 2024. Enhancing llm-based hatred and toxicity detection with meta-toxic knowledge graph. *arXiv preprint arXiv:2412.15268* (2024).
- [38] Hao Zhou, Pei Ke, Zheng Zhang, Yuxian Gu, Yinhe Zheng, Chujie Zheng, Yida Wang, Chen Henry Wu, Hao Sun, Xiaocong Yang, et al. 2021. Eva: An open-domain chinese dialogue system with large-scale generative pre-training. *arXiv preprint arXiv:2108.01547* (2021).
- [39] Xianbing Zhou, Yang Yong, Xiaochao Fan, Ge Ren, Yunfeng Song, Yufeng Diao, Liang Yang, and Hongfei Lin. 2021. Hate speech detection based on sentiment knowledge sharing. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. 7158–7166.

A LLM-Guided Triple Completion

To enrich the Variant-KG with high-quality variant-canonical word pairs, we design a prompt-based strategy to guide large language models (LLMs) in completing missing triples. The prompt defines relation types (e.g., phonetic, visual, semantic) and provides a few-shot template to help the model generate relation-consistent variants for a given canonical term. As illustrated in Figure 8a and Figure 8b, the prompt includes clear instructions and representative examples in both Chinese and English, enabling the model to produce structured outputs in the form of (variant, relation_type, canonical_term). These completions are subsequently filtered and validated to ensure both accuracy and linguistic plausibility.

B Prompt with Knowledge Graph

This section introduces how prompts are augmented with graph-structured evidence retrieved from Variant-KG. These prompts are injected into LLMs to improve the model’s structural awareness and enable more accurate variant classification. As shown in Figure 8c and Figure 8d, each prompt includes neighborhood triples (variant, relation, canonical term) with relation confidence, guiding the LLM to output standardized judgment sentences in a clear and interpretable format.

C Think-Search-Generate Inference

We detail the multi-stage inference pipeline designed under the Think-Search-Generate (TSG) paradigm. Specifically, we demonstrate how the model first hypothesizes candidate variants (**Think**), retrieves structural paths or support evidence from the Variant-KG (**Search**), and finally performs inference with reasoning steps to generate explanatory outputs (**Generate**). Prompt examples and generation outputs are illustrated in Figure 8e and Figure 8f.

D Dataset Construction

We construct a custom dataset, **NetSafety-10K**, to train and evaluate variant-aware toxic content detection models. The dataset is derived from real-world Chinese user-generated content collected across major online platforms, including bullet chats, short comments, and social media posts, ensuring temporal and topical diversity across domains such as entertainment, gaming, politics, and illicit marketing.

All data were strictly anonymized to remove personally identifiable information and sensitive metadata before annotation. The collection and processing fully comply with platform terms and relevant data privacy regulations. To capture disguised toxic language, we applied a multi-step rule-based mining pipeline that detects phonetic substitutions (e.g., pinyin abbreviations), visual perturbations (e.g., lookalike characters or symbol insertions), and euphemistic expressions, generating a comprehensive pool of canonical-variant term pairs.

Each sample was manually annotated by three trained native Chinese speakers, with labels covering toxicity (toxic or benign), violation category (10 predefined classes), and canonical mapping (for disguised forms). Disagreements were resolved by majority voting, and difficult cases were reviewed by an expert annotator.

E Additional Experimental Details

We implement our framework on the **NetSafety-10K** dataset using a graph-augmented prompting strategy, where each prompt integrates variant triples ((variant, relation, canonical)) and reasoning paths retrieved from the Variant-KG. The maximum prompt length is 1024 tokens, and decoding temperature is set to 0.4 for graph construction and 0.1 for reasoning and detection.

For parameter-efficient adaptation, we fine-tune the open-source backbones (Llama-3-8B-Instruct, Qwen3-8B, and Baichuan-7B) using **LoRA** (targeting the $q/k/v/o$ projection matrices; $r = 8$, dropout = 0.1); GPT-4o Mini remains frozen.

The initial Variant-KG contains 12.5K rule-based triples. LLM-based expansion adds 18.7K candidates (confidence > 0.7), and after smoothing and pruning, 22.3K high-confidence triples are

retained across four relation types: phonetic, visual, semantic, and euphemistic variants. Thresholds for addition, pruning, and write-back are 0.7, 0.3, and 0.75, respectively. Training uses AdamW with macro-F1, accuracy, recall, and FPR as metrics.

F Case Study

This section presents qualitative examples comparing different model outputs under various prompting and graph conditions. We show examples where the graph-enhanced prompt significantly improves variant normalization, and contrast them with failure cases caused by ambiguous expressions.

As shown in Figure 7a, baseline LLMs often fail to detect abusive or drug-related content when euphemisms or phonetic variants are used. In contrast, Variant-KG leverages phonetic and metaphorical links to uncover hidden toxicity. For instance, in Figure 7a, a seemingly benign phrase is taken literally by the LLM, while Variant-KG correctly identifies it as a profane variant through phonetic similarity.

Figure 7b further illustrates how vulgar or pornographic content can be disguised through memes or nonsensical wording. Such expressions often evade detection due to contextual ambiguity. With visual and semantic cues, Variant-KG successfully reconstructs the underlying meaning. Overall, graph-enhanced reasoning improves recall and interpretability by revealing implicit links to toxic intent.

G Performance and Robustness Analysis

As illustrated in Figure 6, across four representative violation categories - Political Sensitivity, Vulgar / Pornographic Content, Public Disorder, and Illegal Activity—knowledge graph-enhanced models consistently outperform baseline LLMs across **Accuracy**, **Macro-F1**, and **Recall**. The most pronounced gains appear in Macro-F1 and Recall, suggesting that injecting structured variant knowledge improves the model’s ability to recognize rare, obfuscated, or previously unseen toxic expressions rather than merely boosting easy majority cases. Notably, the improvement is especially evident in Political Sensitivity and Illegal Activity, where KG-enhanced models yield substantially higher Recall than non-enhanced counterparts, reflecting stronger resilience to adversarial perturbations such as homophones, phonetic disguises, and symbol insertions. Overall, these radar charts indicate that Variant-KG provides broad, category-level robustness rather than isolated gains on a single violation type.

Robustness Gains from Prequential KG Expansion. To further analyze robustness under evolving language, we adopt a test-then-write schedule for continual graph evolution (see Figure 5). The evaluation stream is partitioned into rounds $\{S_r\}$, and at each round r we freeze the backbone, prompts, fusion weights, and thresholds, evaluate on S_r using the current graph KG_{r-1} , and then write back only high-confidence, quality-checked variants to obtain KG_r . This prequential protocol prevents train-test leakage and isolates the effect of graph growth: performance changes across rounds can be attributed primarily to newly incorporated variant knowledge rather than parameter updates. Empirically, Figure 5 shows a steady, monotonic improvement across rounds with noticeably reduced variance, indicating that controlled write-back yields smoother adaptation under distribution shift.

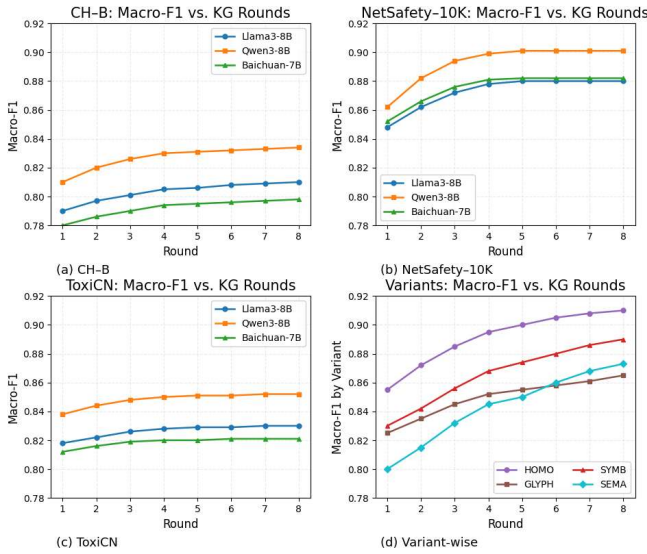


Figure 5: Effect of prequential KG expansion on robustness. The figure reports round-wise Macro-F1 under the test-then-write schedule, showing steady improvements across evaluation rounds and more stable performance as the KG grows.

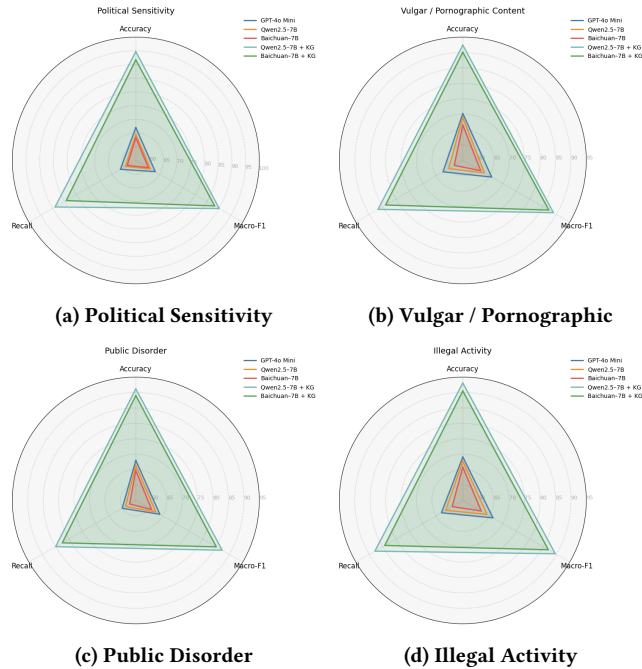
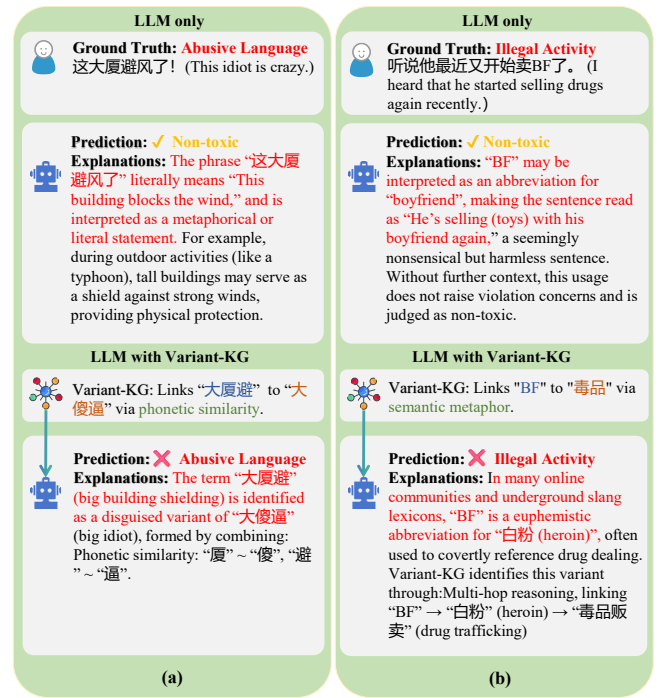
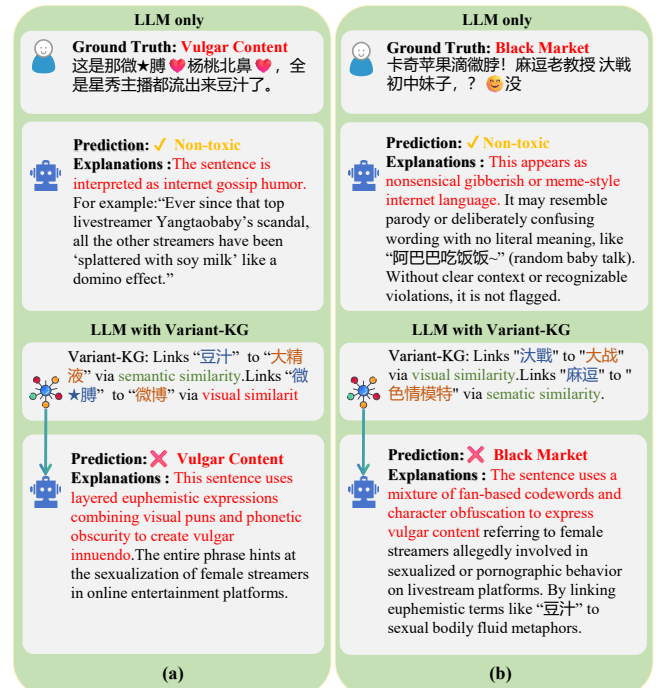


Figure 6: Radar chart comparison across four violation categories. KG-enhanced models consistently outperform vanilla LLMs in Accuracy, Macro-F1, and Recall.



(a) Case Study 1: Resolving covert abuse and drug-related variants.



(b) Case Study 2: Uncovering vulgarity and covert advertising.

Figure 7: Case study visualizations illustrating how Variant-KG resolves disguised toxic expressions across multiple domains.



Figure 8: Chinese and English prompt templates used in Variant-KG construction and TSG reasoning. Each pair illustrates the original prompt (left) and its translated or augmented version (right).